TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	91.4 ± 0.66
Correlation Stability	94.8 ± 0.87
Distribution Consistency	94.4 ± 1.11
Entropy Preservation	95.2 ± 0.75
Mean Z Score Drift	0.0 ± 0.01
Categorical Fidelity	98.5 ± 0.67
Privacy Risk	4.1 ± 1.22
Data Density	100.0 ± 0.00
Coverage Score	59.0 ± 2.53
Feature Accuracy	94.6 ± 1.80
Novelty Score	0.8 ± 0.01

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	44.863	+0.117	0.3%	OK
TB	3.299	3.454	+0.155	4.7%	OK
DB	1.486	1.580	+0.094	6.3%	OK
Alkphos	290.576	277.052	-13.525	4.7%	OK
Sgpt	80.714	76.646	-4.068	5.0%	OK
Sgot	109.911	91.468	-18.443	16.8%	Warn
TP	6.483	6.469	-0.014	0.2%	OK
ALB	3.142	3.168	+0.026	0.8%	OK
A/G Ratio	0.947	0.957	+0.010	1.0%	OK
Selector	1.286	1.249	-0.038	2.9%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	91.4 ± 0.66
Correlation Stability (*2)	94.8 ± 0.87
Distribution Consistency (*3)	94.4 ± 1.11
Entropy Preservation (*4)	95.2 ± 0.75
Mean Z Score Drift (*5)	0.0 ± 0.01
Categorical Fidelity (*6)	98.5 ± 0.67
Privacy Risk (*7)	4.1 ± 1.22
Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	59.0 ± 2.53

Feature Accuracy (*10)	94.6 ± 1.80
Novelty Score (*11)	0.8 ± 0.01

[1] (*12) Algorithm: Contextual constraint or model parameter.

[2] (*13) Folds: Contextual constraint or model parameter.

[3] (*14) Context %: Contextual constraint or model parameter.

[4] (*15) Generation %: Contextual constraint or model parameter.

[5] (*16) Join context: Contextual constraint or model parameter.

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6615384615384616	0.6658119658119658	0.6564102564102564	-0.00512820512820513	
F1	0.7638710786531677	0.7767390355874583	0.7638404506325219	-3.062802064587311e-05	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6820512820512821	0.6923076923076923	0.6991452991452991	0.0170940170940171	
F1	0.7893107648024189	0.7990098428181727	0.8027771740181592	0.013466409215740448	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6777777777777778	0.6863247863247863	0.6760683760683761	-0.0017094017094016977	
F1	0.7863567503178366	0.7935363033732606	0.7672844296671923	-0.01907232065064408	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.558974358974359	0.5512820512820513	0.5188034188034187	-0.04017094017094018	*
F1	0.5587432119380917	0.5562611019567143	0.4955179067012244	-0.06322530523686724	*

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.7025641025641025	0.7136752136752136	0.02735042735042733	
F1	0.7971299319768493	0.8141211623388974	0.8176844514774848	0.020554519500635426	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6965811965811965	0.7145299145299145	0.7102564102564102	0.01367521367521368	
F1	0.803778911401206	0.8203861112086258	0.8164240964908336	0.012645185089627631	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7059829059829059	0.6991452991452991	0.7017094017094017	-0.00427350427350427	
F1	0.8128002697439015	0.8154338978327195	0.8155945274218498	0.002794257677948186 4	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	0.6803418803418804	0.7059829059829059	0.003418803418803418	
F1	0.7997903571549948	0.7885116465747843	0.8039375408750338	0.004147183720038994	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.7068376068376068	0.7068376068376068	0	
F1	0.8279091624529537	0.8279091624529537	0.8279091624529537	0	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.7179487179487178	0.7051282051282051	0.03076923076923077	
F1	0.7821343746683586	0.8192635048403691	0.8103033371957009	0.028168962527342155	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	66.0	Female	0.8	0.2	194.0	32.0	25.0	6.4	3.0	0.9
2	35.0	Male	1.8	0.3	320.0	36.0	34.0	7.1	3.7	0.8856348517470326
3	58.0	Male	30.5	14.103017278904247	751.0	23.0	36.0	5.5	2.5	0.7946159179079746
4	32.0	Female	0.8	0.2	195.0	18.0	14.0	8.36373438139097	4.8	1.3
5	60.0	Male	2.8	1.3	238.0	53.0	66.0	5.1	1.8	0.4635611956197385
6	72.0	Male	15.124853468922652	6.377910371195708	234.0	35.0	47.0	5.2	2.0	0.8
7	33.0	Male	7.1	3.7	196.0	622.0	497.0	6.9	3.6	1.09
8	38.0	Male	2.2	1.0	310.0	119.0	42.0	7.9	4.1	1.0
9	64.0	Male	3.0	1.4	248.0	46.0	40.0	6.5	3.2	0.9
10	40.0	Female	0.9	0.2	285.0	32.0	27.0	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	38.0	Male	0.7	0.2	180.0	18.0	21.0	6.3	4.0	1.4
2	47.0	Male	1.0	0.2	158.0	23.0	41.0	5.9	2.6	1.0
3	53.0	Male	6.2937094434003145	2.5097113937494555	237.0	27.0	26.0	7.4	3.5	0.9
4	25.0	Male	1.4	0.4	188.0	138.0	126.0	5.9	2.8	0.9
5	32.0	Female	0.6	0.1	145.0	37.0	28.0	6.5	3.5	1.1
6	60.0	Male	1.7	0.2	292.0	24.0	14.0	8.3	3.9	0.6
7	50.0	Female	1.0	0.3	191.0	22.0	31.0	7.8	4.0	1.0
8	52.0	Female	0.6	0.1	194.0	10.0	12.0	6.9	3.3	0.9
9	30.0	Male	0.8	0.2	182.0	46.0	57.0	7.8	4.3	1.2
10	58.0	Male	0.8	0.2	298.0	33.0	59.0	6.2	3.1	1.0

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	60.0	Male	1.0	0.5	188.0	31.0	74.0	6.1	2.6	0.8
2	36.0	Female	0.9	0.2	310.0	60.0	56.0	5.6	2.5	0.6
3	51.0	Male	1.1	0.2	514.0	232.0	849.0	4.434619699721029	2.0	0.8
4	60.0	Male	18.652822638903501	12.378332180683037	272.0	42.0	53.0	4.3	1.8	0.8
5	68.0	Male	22.861332754518084	12.778717380750493	279.0	74.0	180.0	6.1	2.3	0.5379275748439026
6	72.0	Male	1.5	0.8	202.0	25.0	28.0	6.2	3.7	1.1
7	44.0	Female	1.9	0.6	298.0	378.0	602.0	6.6	3.3	1.0
8	30.0	Male	0.8	0.2	182.0	46.0	57.0	7.8	4.3	1.2
9	53.0	Male	0.9	0.4	238.0	17.0	14.0	6.6	2.9	0.8
10	28.0	Male	0.8	0.3	190.0	20.0	14.0	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	72.0	Male	1.7	0.5	177.0	31.0	41.0	5.100302469261049	2.2	0.9
2	42.0	Male	9.35497978364304	4.020610321791184	271.0	53.0	42.0	7.2	3.9	1.0
3	18.0	Male	0.7	0.2	235.0	28.0	58.0	6.2	2.5	0.6
4	62.0	Male	1.0	0.3	186.0	46.0	36.0	7.2	3.5	1.0
5	18.0	Male	0.8	0.4	117.0	35.0	47.0	6.4	4.1	1.85
6	33.0	Male	2.2	3.2	293.0	25.0	27.0	7.9	4.048509132603971	0.96
7	42.0	Male	0.7	0.2	197.0	64.0	33.0	5.8	2.4	0.7
8	69.0	Male	0.9	0.2	215.0	32.0	24.0	6.9	3.0	0.7
9	53.0	Male	1.6	0.9	178.0	44.0	59.0	6.5	3.9	1.5
10	49.0	Male	1.3	0.4	206.0	30.0	25.0	6.0	3.1	1.06

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	66.0	Female	1.1	0.2	1544.0	49.0	59.0	7.3	3.2	0.7
2	61.0	Male	1.0	0.7	114.0	48.0	33.0	6.2	3.7	1.4741760437887195
3	60.0	Male	0.8	0.2	205.0	58.0	48.0	8.5	4.5	1.0
4	60.0	Female	1.4308693325053183	0.6	244.0	42.0	44.0	6.1	2.0	0.4
5	54.0	Male	3.286072772106263	1.9	189.0	22.0	19.0	6.8	3.6	1.1
6	60.0	Female	6.787912758047434	4.1129753796285977	358.0	66.0	65.0	5.9	3.0	0.9
7	33.0	Male	2.1	0.7	205.0	50.0	38.0	6.8	3.0	0.7
8	22.0	Female	6.7	3.2	850.0	154.0	248.0	6.2	2.8	0.8
9	48.0	Male	5.8	2.5	802.0	133.0	88.0	6.0	2.8	0.8
10	84.0	Female	0.7	0.2	188.0	13.0	21.0	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	28.0	Male	6.8	1.6	158.0	50.0	60.0	7.4	4.0	1.0
2	30.0	Male	0.8	0.2	320.0	20.0	20.0	5.7	2.5	0.7
3	40.0	Male	1.1	0.2	488.0	26.0	29.0	5.113629273543566	2.5	1.1
4	18.0	Male	0.9	0.2	215.0	27.0	30.0	8.190972628380745	4.0	0.7
5	40.0	Male	0.7	0.2	188.0	32.0	20.0	7.032654886932977	4.672938675071384	2.5
6	37.0	Female	3.0	1.6	208.0	30.0	30.0	5.1	3.0	1.1
7	50.0	Male	0.9	0.3	194.0	190.0	73.0	7.5	3.9	1.0
8	49.0	Male	0.6	0.1	218.0	50.0	53.0	5.0	2.4	0.9
9	18.0	Male	0.8	0.2	228.0	55.0	54.0	6.9	4.0	1.3
10	30.0	Female	0.7	0.2	194.0	32.0	36.0	7.5	3.6	0.92

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	31.0	Female	0.7	0.2	203.0	80.0	104.0	8.1	4.1	1.0
2	22.0	Male	3.4120797680990393	0.8	185.0	178.0	111.0	8.0	4.4	1.4
3	12.0	Male	4.2	1.4268066184249602	252.0	27.0	43.0	6.2	3.2	1.1
4	25.0	Male	0.8	0.2	198.0	22.0	25.0	7.1	3.5	0.7
5	40.0	Male	0.7	0.2	172.0	21.0	20.0	5.5	3.1	1.3
6	46.0	Male	0.6	0.2	165.0	20.0	40.0	7.2	3.4	0.9
7	13.0	Female	0.7	0.2	350.0	17.0	24.0	7.4	4.0	1.1
8	54.0	Female	23.2	12.6	574.0	43.0	47.0	7.2	3.5	0.9
9	63.0	Male	0.9	0.2	194.0	52.0	45.0	6.0	3.9	1.85
10	46.0	Male	0.6	0.2	290.0	26.0	21.0	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	18.0	Male	0.8	0.2	205.0	28.0	34.0	5.5	2.5	1.0
2	34.0	Female	0.6	0.1	482.0	114.0	51.0	6.0	2.5	0.6
3	49.0	Male	26.885224497371073	11.983439383067752	1673.0	122.0	500.0	5.1	1.6	0.3650098350600507
4	32.0	Female	0.7	0.2	134.0	15.0	14.0	7.1	4.1	1.0
5	70.0	Male	1.5	0.5	349.0	35.0	25.0	5.3	2.5	0.7
6	60.0	Male	2.9	0.9	245.0	32.0	42.0	5.1	1.8998934387490982	0.8
7	48.0	Male	0.7	0.2	208.0	15.0	30.0	4.6	2.1	0.8
8	60.0	Male	22.8	12.6	962.0	53.0	41.0	6.9	3.3	0.9
9	27.0	Male	1.0	0.3	180.0	56.0	111.0	6.8	3.9	1.85
10	28.0	Female	0.8	0.2	309.0	55.0	23.0	6.8	4.1	1.51

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	46.0	Male	1.3	1.2	196.0	58.0	61.0	7.793285137656028	3.3	0.8
2	45.0	Male	0.6	0.2	150.0	132.0	97.0	5.1	3.0	1.0
3	69.0	Male	15.770454992723012	10.187181868939819	239.0	117.0	92.0	7.9876062074363405	3.2	0.5
4	38.0	Male	0.9	1.2	350.0	25.0	66.0	8.2	4.0	1.0
5	62.0	Female	1.7747865585771365	1.3	293.0	17.0	42.0	5.2	2.1	0.7
6	13.0	Male	0.7	0.1	239.0	31.0	21.0	6.8	3.4	1.0
7	26.0	Male	2.0	0.9	195.0	24.0	65.0	7.8	4.3	1.2
8	47.0	Male	2.7	1.3	275.0	123.0	73.0	6.2	3.3	1.1
9	74.0	Male	1.0	0.3	175.0	30.0	32.0	6.4	3.4	1.1
10	48.0	Male	3.2	1.6	257.0	33.0	116.0	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	45.0	Male	2.2	1.3243197105649416	352.0	91.0	56.0	6.6	2.9	0.8
2	47.0	Female	1.0	2.2547993244574966	172.0	16.0	21.0	8.2	4.3	1.5059363650763913
3	42.0	Female	0.8	0.2	381.0	42.0	141.0	7.0	3.3	0.8
4	60.0	Male	1.8	0.5	196.0	25.0	74.0	3.6	0.9717218647782875	0.4700191463096382
5	9.0	Male	0.7363611018833026	0.2172108519922375 2	185.0	72.0	235.0	5.9	2.6	1.0
6	45.0	Male	0.8	0.2	300.0	33.0	52.0	6.9	3.0	0.7
7	54.0	Female	5.5	3.2	350.0	67.0	42.0	7.0	3.2	0.8
8	43.0	Male	22.5	11.8	143.0	22.0	143.0	6.6	2.1	0.46
9	57.0	Male	4.5	2.3	315.0	120.0	105.0	7.0	4.0	1.3
10	65.0	Male	0.7	0.2	199.0	19.0	22.0	6.3	3.6	1.3